

java script basics

- java script is a **client side scripting language**
- scripting languages are almost equivalent to **light weight** programming language
- java script is **case sensitive**

java script supported entity

- data types
- variables
- functions
- objects
- arrays
- constructor
- conditional statements
 - if
 - else if
 - else
 - switch
- loop
 - for loop
 - while loop
 - do while loop
 - for/in

operators supported in JS

< <= > >= == ===
& | && || + - * /
+= -= ++ -- *= != %

JavaScript doesn't support

- classes
- interface
- enum
- structures

where to write JavaScript??

JavaScript code can be written inside html file or in JavaScript file



javascript code written in html file is called as **inline javascript**



javascript code written in external javascript file is called as **external javascript**

writing JavaScript in html file

- to write JavaScript code in html file you have to use **<script>** tag.

we can write JavaScript code

within <head> tag

within
<body> tag

after </body>tag

```
<!DOCTYPE html>
<html>
<head>
<script type="text/javascript">

</script>
</head>
<body>

<script type="text/javascript">

</script>
</body>

<script type="text/javascript">

</script>

</html>
```

variables in JavaScript

- using variables we can store data

declaring variables:

we can declare variables **with** or **without** using **var** keyword

```
var v=10;  
    a=9.8;  
    b=1000;  
    c='ABC';
```

```
var d="abc";  
var e=objects;  
f=false;  
g=functions;
```

- JavaScript is a **loosely typed** language

(data type is not required while declaring variable)

ex : **var v1=10; or v1=10 ;**

strictly typed language/strongly typed language

(data type is required while declaring variable)

ex : **in c# | java : int i=10 ;**

functions in JavaScript

using functions we can perform some actions/work

syntax:

C# | java sample method

<script>

Inside script Tag

```
function function_name(v1,v2,v3....)
{
  ---JavaScript code---
}
```

</script>

Where JavaScript functions are written?

Inside class

return type

```
public int function_name(v1,v2,v3....)
{
  ---Java | C# code---
}
```

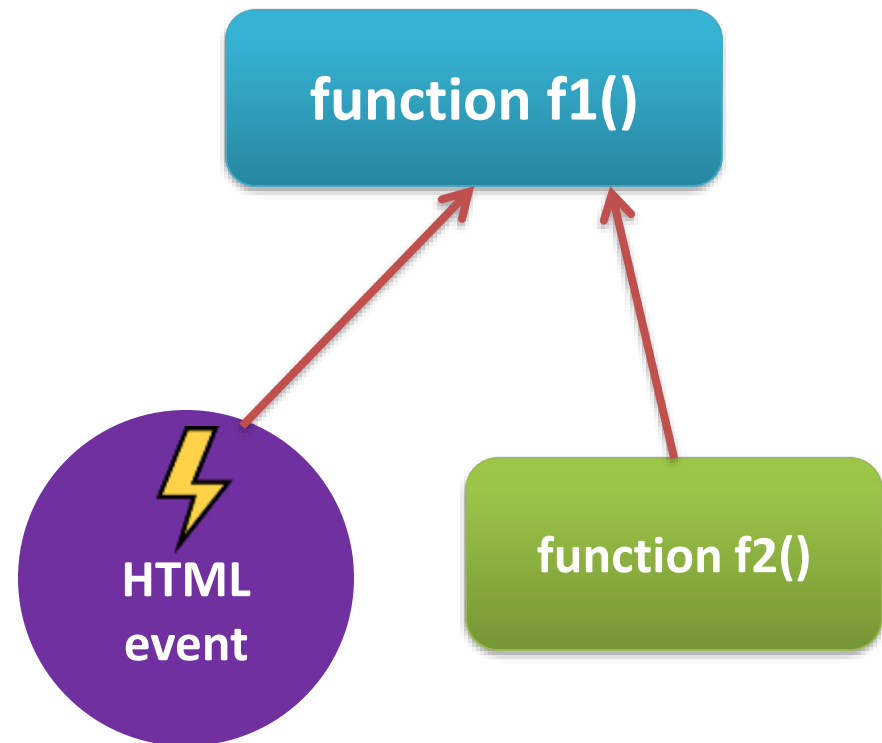
Where c#/java methods are written?

JavaScript function declaration will **not support return type** but we can return values using return keyword.

JavaScript will **not support access modifiers**

calling functions

- we can call a function from another function or from an html event.
- html event is a code executed for user actions.



js sample1



how will you display output in JavaScript ?

```
alert(message)
document.write(text)
```

req:1.when the user **clicks on submit1** display **'hello'**

2.when the user **moves mouse on top of submit2**: display **'hi'**

```
<html>
<head>
  <title>My Page</title>
  <script type="text/javascript">
    function f1()
    {
      alert('HELLO');
    }
    function f2()
    {
      f3();
    }
    function f3() {
      document.write('HI')
    }
  </script>
</head>
<body>
  <input type="submit" id="btn1" value="submit1" onclick="f1()" />
  <input type="submit" id="btn2" value="submit2" onmouseover="f2()" />
</body>
</html>
```

sample.html

SUBMIT1 SUBMIT2

onclick
ondblclick
onkeydown
onkeypress
onkeyup
onmousedown
onmousemove
onmouseout
onmouseover

calling one function

er function

which event will be fired when user clicks
On a button?

calling function with an event

TYPES OF VARIABLES

```
graph TD; A[TYPES OF VARIABLES] --> B[Local variables]; A --> C[Global variables]
```

Local variables

Global variables

Local variables : variables declared inside the function

Global variables : **variables** declared **outside** the **function** or variables declared **in function body** without **var** keyword.

local & global variables

```
<script type="text/javascript">
```

```
var v1 = 10;
```

```
v2 = "palle";
```

→ v1,v2 are global variables

```
function fn1(v3, v4)
```

→ v3,v4 are parameters/local variables

```
{
```

```
var v5 = "palle university";
```

→ v5 is local variable

```
v1 = v1 + 6;
```

→ we can call global variable inside a function

```
v6 = "hello";
```

→ v6 is global variable as we are not using var keyword

```
}
```

```
function fn2()
```

```
{
```

```
alert(v6);
```

→ v6 is global variable we can use anywhere

```
alert(v5);
```

✗ → v5 is local variable we cannot use in another function

```
}
```

```
</script>
```

Conditional statements in JavaScript

- Java script is a light weight programming language
- Java script supports conditional statements
 1. if ,else if and else statements .
 2. switch statements(faster compared to multiple else if statements)

Note:

All conditional statements (if ,else if ,switch) are written in function body

Syntax for if else in JavaScript

```
if(exp1) {  
  /*----js code----*/  
}
```

→ If exp1 is true enters into body
or goto else if

```
else if(exp2) {  
  /*----js code----*/  
}
```

→ If exp2 is true enters into body
or goto next else if

```
else if(exp3) {  
  /*----js code----*/  
}
```

→ If exp3 is true enters into body
or goto else

```
else {  
  /*----js code----*/  
}
```

→ Only one else statement is
possible for if

Note: we can write 0 or more else if statements
We can write 0 or 1 else statement.

Using multiple expressions in conditional statements

If(exp1 && exp2) **If exp1 and exp2 both are true, execute the body**

{

----js code----

}



Use conditional and (&&) operator

If(exp1 || exp2) **If any one of the expression is true ,execute the body**

{

----js code----

}



Use conditional or(||) operator

Syntax for switch statement in javascript

```
switch(variable/expression)
{
  case  :
    /*jsgcode*/
    break;

  case  :
    /*jsgcode*/
    break;

  case  :
    /*jsgcode*/
    break;

  default:
    /*jsgcode*/
    break;
}
```

 Predicted value for variable

Assignment on conditional Statements:-

DisplayDayname.html



On click: "fndisplaydayname(3)"

function fndisplaydayname(dno)

{

1 → Monday Invalid day

2 → Tuesday

3 → Wednesday

4 → Thursday

5 → Friday

6 → Saturday

7 → Sunday

}

if else statement assignment solution

```
<html>
<head>
  <title></title>
  <script type="text/javascript" >
    function fndisplaydayname(dno) {
      if (dno == 1) {
        alert("monday");
      }
      else if (dno == 2) {
        alert("tuesday");
      }
      else if (dno == 3) {
        alert("wednesday");
      }
      else if (dno == 4) {
        alert("thursday");
      }
      else if (dno == 5) {
        alert("friday");
      }
      else if (dno == 6) {
        alert("saturday");
      }
      else if (dno == 7) {
        alert("sunday");
      }
      else {
        alert("invalid day");
      }
    }
  </script>
</head>
<body>
<input type="submit" id="btnsubmit" value="displaydayname" onclick="fndisplaydayname(3)" />
</body>
</html>
```


Assignment solution using switch

```
<html >
<head>
  <title></title>
  <script type="text/javascript">
    function displayday(dno) {
      switch (dno) {
        case 1:
          alert("monday")
          break;
        case 2:
          alert("tuesday")
          break;
        case 3:
          alert("wednesday")
          break;
        case 4:
          alert("thursday")
          break;
        case 5:
          alert("friday")
          break;
        case 6:
          alert("saturday")
          break;
        case 7:
          alert("sunday")
          break;
        default:
          alert("invalid day")
          break;
      }
    }
  </script>
</head>
<body>
<input type="submit" id="tbsubmit" value="displayday" onclick="displayday(3)" />
</body>
</html>
```

ARRAYS

- Using arrays we can store multiple values under a single variable name.
- Usually, to store 3 persons name ,we will declare three variables

```
var name1 = "paul";  
var name2 = "kevin";  
var name3 = "brain";
```



Incorrect approach

Using arrays

```
var names=['paul','kevin','brain'];
```

Array Syntax & Internals

Array syntax: `var array_name=[item1,item2,item3...]`

Array Sample: `var names=['paul','kevin','brain'];`



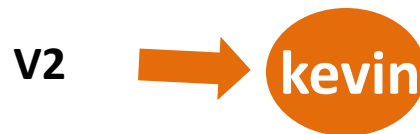
To retrieve 1st cell data

```
var v1 = names[0]
```



To retrieve 2nd cell data

```
var v2 = names[1];
```



To retrieve 3rd cell data

```
var v3 = names[2];
```

v3 = "brain"



to modify 2nd cell data:

`names[1] = "kavitha";` if this code is executed kevin will be replaced with kavitha

To count total number of cells in array: `names.length`

Loops

Javascript supports



- For (;;)

- For/in



Similar to foreach loop In **c#** and **java**

- While()

- do while()

- Loops are useful for eliminating the duplication of code (Or) for executing a set of code for a specific number of times
- In this tutorial we will learn only for(;;) loop

Loop sample

var nos =

101	21	67	278	45	334
-----	----	----	-----	----	-----

Req: Display all items present in the array nos

Index numbers 0 1 2 3 4 5

```
alert (nos[ 0 ]);  
alert (nos[ 1 ]);  
alert (nos[ 2 ]);  
alert (nos[ 3 ]);  
alert (nos[ 4 ]);  
alert (nos[ 5 ]);
```

Using loops , we can reduce the logical code duplication

```
for( i=0 ; i<nos.length ; i++)  
{  
    alert (nos[ i ])  
}
```



Use length property to find the length of array



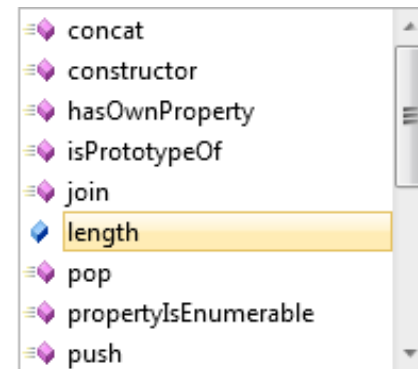
If I add another element
To the array, will it work..?

Current i value

i = 0
i = 1
i = 2
i = 3
i = 4
i = 5
i = 6

Condition

0<=5
1<=5
2<=5
3<=5
4<=5
5<=5
6<=5



67
278
45
334

End of Loop

TEST1

Test must be conducted based on html css and javascript by giving 3 separate questions



Execute all code and do
all Exercises at Home

End of
DAY 1

debugging

- using **debugging** we can see **code execution seq.**
- we can **see run time variables data.**
- we can get **missing intelligence.**

where the JavaScript code are debugged ?

- visual studio
- local server
- web server
- **browser** ✓

debugger keyword

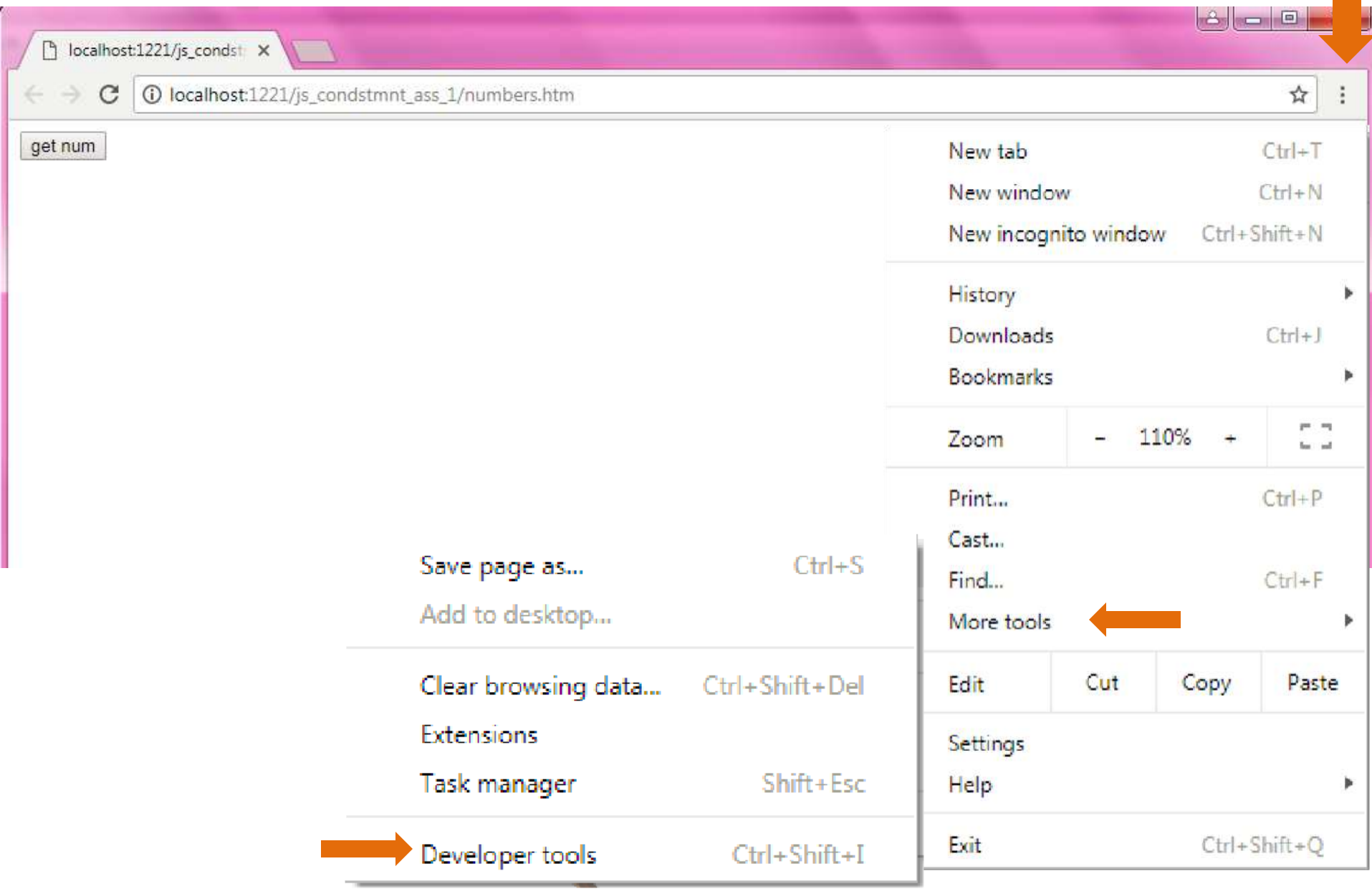
- For debugging javascript code , we have to include **debugger;** command in required funtion.

```
<script type="text/javascript">  
  function f1() {  
    debugger;  
    -----js code-----  
  }  
</script>
```

is it possible to see
received js code ?

yes. right click on
browser , go to view
source.

debugging in chrome – part1



debugging in chrome – part2

F11

click on button to start debugging. use **F10** | **F11** key to navigate

See the runtime variables data
We will get the missing IntelliSense support.

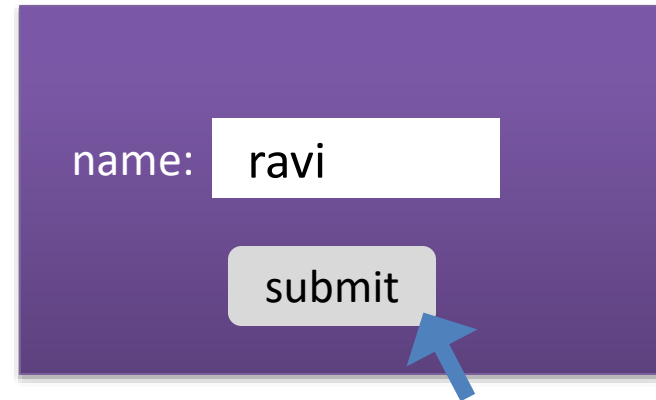
```
1 <!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/xhtml1" >
2
3 <html>
4 <head>
5   <title></title>
6   <script type="text/javascript">
7     fun 10 on f1() {
8       debugger;
9       v1 = input#btn1
10      v1
11    }
12  </script>
13 </head>
14 <body>
15   <input type="text" value="10" />
16   <input type="button" value="getnum" />
17 </body>
18 </html>
```

Debugger Console:

```
v1 = input#btn1
accept: ""
accessKey: ""
align: ""
alt: ""
assignedSlot: null
attributes: NamedNodeMap {0: type, 1: id}
autocapitalize: "none"
autocomplete: ""
autofocus: false
baseURI: "http://localhost:1472/jsass_21"
checked: false
childElementCount: 0
childNodes: NodeList [1]
```

reading controls data

```
function f1()  
{  
→ v1 = document.getElementById("tbname");  
→ v2 = v1.value;  
→ alert(v2);  
→ }
```



A web form with a purple background. It contains a text input field with the label "name:" and the value "ravi". Below the input field is a grey "submit" button. A blue arrow points to the submit button.

`<input type="text" id="tbname"/>`



Execute all code and do
all Exercises at Home

End of
DAY 2

validations

we can verify whether user given data is in valid format or not

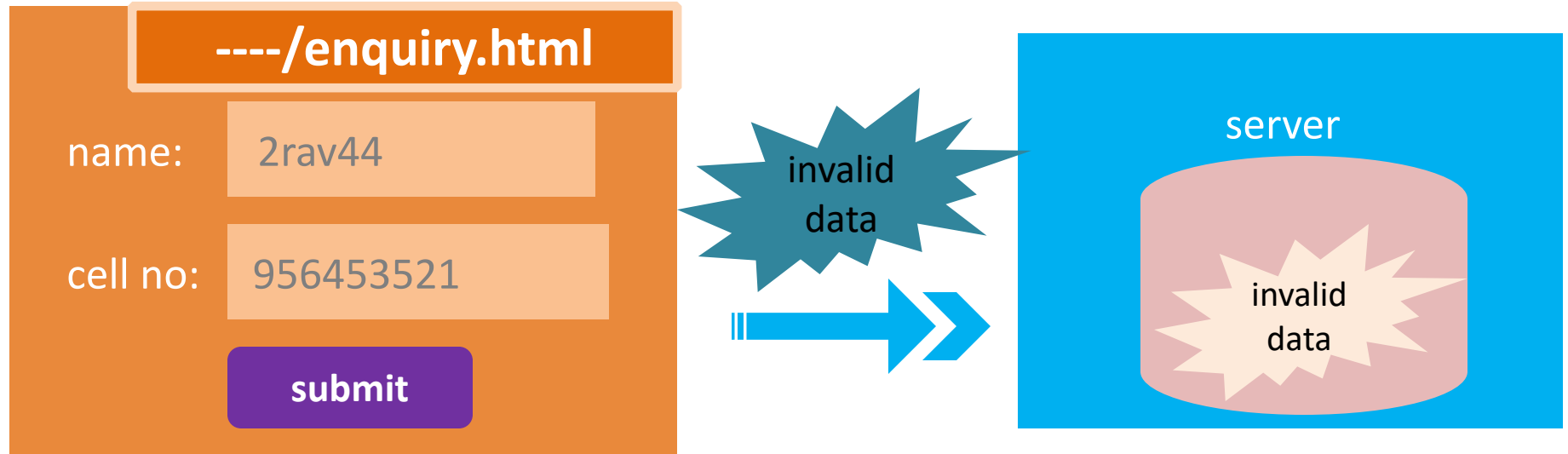


Why should we validate the user given data?

We will understand in the next slide

why to validate a data

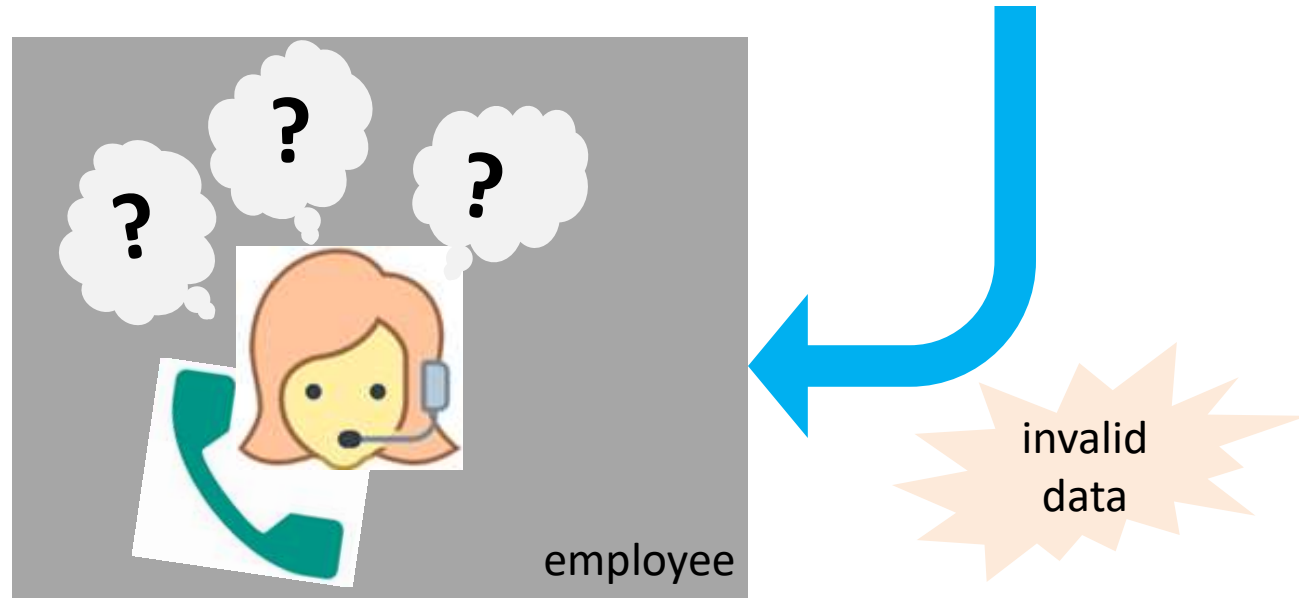
assume there is company having a webpage enquiry.html



as a programmer we must not allow the bad data inside the application



how to find which is good data and bad data?



validation using regular expressions

regular expression:

we can find whether a variable data is present in valid format or not

regular expression:

meta char + **quantifiers** + **brackets**

meta characters

Meta chars	descriptions/meaning
.	Any single character
\w	Used for finding word characters(a-z,A-Z,0-9,_)
\W	Used for finding any non word characters
\d	For finding digits
\D	For finding non digits
\s	White space
\S	Non white space
-----	-----

types of quantifiers

Quantifiers	meanings
(Expression) +	Expression for 1 or more times
(Expression) *	Expression for 0 or more times
(Expression) ?	Expression for 0 or 1 time
(Expression) {no}	Expression for exactly specified number of times
(Expression) {minno,maxno}	Expression specified in that specific range

brackets

Brackets[]	examples
[set of characters]	[a b h l]
[range of characters]	[d-k] means any character between d to k
[Checking special character]	Any special character used in regular expression must be placed in square brackets

Sample validations

```
var v1 = '909089790';
```

Req: To check whether v1 has 10 digits or not

Regular expression : `\d{10}`

*Regular expression is written in between

`/^_____$/`

```
var rex1 = /^ \d{10} $/;
```

match function

- **match** function is used for validations.
- `variable_which_holds_data.match(variable_holding_regular_expression);`

returns null if variable data not matching to regular expression

```
if(v.match(rex1)==null)
{
    alert("error message");
}
```

returns null if variable data not matching to regular expression

```
if(!v.match(rex1))
{
    alert("error message");
}
```

Validation Lab-1

- write validation expression for checking a variable named “v” is having + or – symbol exactly one time between one or more english alphabets.
- write validation expression for indian cell phone number i.e (+91-XXXXXXXXXX) or without +91-

Sol:var v2=/^[+]*[91]*[-]*\d{10}\$/;

Validation Sample2

- Req:Name must contain minimum 3 alphabets ,maximum can be 10



A web form titled "A.Html" with an orange background. It contains a label "name:" followed by a white input field. Below the input field is a blue "submit" button.

Regular expression: `/^[a-z A-Z]{3,10}$/`

Validation Sample3

- Show validation expression for +91-10digit mob num
- validation for email id
- `var v3 = /^\\w+([\\.-]?\\w+)*@\\w+([\\.-]?\\w+)*([\\.]?\\w)+$/`
- validation for full name with a space in between



Execute all code and do
all Exercises at Home

End of
DAY 3